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Complete Configuration Guide for the C-DATA OLT: From Initial Setup to Service Deployment

The following sections provide the commands required to configure a new C-DATA FD1616S-B1 OLT and explain the function of each command.

The following example IP addresses are used:

Local OLT IP: 192.168.1.100/24
Laptop IP: 192.168.1.10/24

Production Management VLAN: 112
New OLT IP: 10.49.12.222/30
New Gateway: 10.49.12.221
Retail uplink: XGE 0/0/1
Test customer VLAN: 501
Test GPON port: GPON 1

Caution: The IP addresses 10.49.12.222/30 and 10.49.12.221 are used for demonstration purposes only. Use the actual IP addresses assigned to your network.

Step 1: Connect the Laptop to the OLT

Physical connection:

Laptop Ethernet Port
|
| LAN Cable
▼
OLT MGMT Port

Configure the following on the laptop:

IP address: 192.168.1.10
Subnet mask: 255.255.255.0
Default gateway: N/A
DNS: N/A

Test from CMD: ping 192.168.1.100

What does it do? It checks whether the OLT at 192.168.1.100 is reachable from the laptop.

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If you receive a reply, connect through SSH: ssh username@192.168.1.100

Alternatively, open a browser: http://192.168.1.100

Step 2: Configuration mode

After logging in to the OLT:

```
# enable
# configure terminal
```

On some firmware versions:

```
# enable
# config
```

Command	Function
enable	Enters administrator/privileged mode
configure terminal	Enters configuration mode
config	Short form of the configuration-mode command on some firmware versions

Step 3: OLT hostname

Command:

```
(config)# sysname Ambarkhana_GPON-01
```

Command Explanation:

Sets the OLT hostname. This name appears in monitoring systems, SSH sessions, and configuration backups.

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Step 4: Local MGMT IP

Command:

```
Ambarkhana_GPON-01(config)# interface mgmt
Ambarkhana_GPON-01(config-interface-mgmt)# ip address 192.168.1.100 24
Ambarkhana_GPON-01(config-interface-mgmt)# exit
```

Command Explanation:

Command	Function
interface mgmt	Enters the dedicated management-port configuration mode
ip address 192.168.1.100 24	Assigns 192.168.1.100/24 to the MGMT port
exit	Exits interface configuration mode

Here, /24 = 255.255.255.0

The laptop and OLT are on the same network:

Laptop: 192.168.1.10/24
OLT: 192.168.1.100/24

10.49.12.222/30 and 10.49.12.221 belong to the same 10.49.12.220/30 network.

Step 5: Create a New Administrator

Command:

```
Ambarkhana_GPON-01(config)# user name noc password ******** privilege-level 15
```

Command Explanation:

Parameter	Meaning
user name noc	Creates a user named noc
password	Sets the login password
privilege-level 15	Assigns the highest administrator privilege level

Step 6: Management services

Command:

```
Ambarkhana_GPON-01(config)# service ssh enable
Ambarkhana_GPON-01(config)# service https enable
Ambarkhana_GPON-01(config)# service snmp enable
Ambarkhana_GPON-01(config)# service http enable
```

Command Explanation:

Command	Function
service ssh enable	Enables secure remote CLI access
service https enable	Enables secure web management
service snmp enable	Enables NMS monitoring
service telnet enable	Enables remote CLI access to the OLT over the network
service http enable	Enables standard web management

Use SSH/HTTPS in a production network. Do not enable Telnet unless it is required.

Step 7: Create VLANs

Complete VLAN list for the OLT:

vlan 1,44,86,110,112,200,221,257,410-411,420
vlan 431,480,501-503,505-508,785-787,934,1120

Command:

```
Ambarkhana_GPON-01(config)# vlan 1121-1124

Create vlan successfully:

1121-1124
```

Step 8: Production management IP

Assume the NOC team has assigned:

Management VLAN: 112
OLT IP: 10.49.12.222/30
Gateway: 10.49.12.221

Command:

```
Ambarkhana_GPON-01(config)# interface vlanif 112
Ambarkhana_GPON-01(config-vlanif-112)# ip address 10.49.12.222 30
Ambarkhana_GPON-01(config-vlanif-112)# exit
```

Command Explanation:

Command	Function
interface vlanif 112	Creates or opens the Layer 3 interface for VLAN 112